isolated process. In order to indicate that it never decreases, we write:

$$\Delta S \geq 0$$

This reads: “Delta *S* is always greater than or equal to zero,” and we call this “the second principle of thermodynamics” (the first being the conservation of energy). Its nub is the fact that heat passes only from hot bodies to cold, never the other way around.

Forgive me for the equation—it’s the only one in the book. It is the equation for time’s arrow, and I could hardly refrain from including it in my book about time.

It is the only equation of fundamental physics that knows any difference between past and future. The only one that speaks of the flowing of time. Behind this unusual equation, an entire world lies hidden.

Revealing it will fall to an unfortunate and engaging Austrian, the grandson of a watchmaker, a tragic and romantic figure, Ludwig Boltzmann.

BLUR

It is Boltzmann who begins to see what lies behind the equation $\Delta S \geq 0$, throwing us into one of our most dizzying dives toward understanding the intimate grammar of our world.



Boltzmann works in Graz, Heidelberg, Berlin, Vienna, and then in Graz again. He liked to attribute his restlessness to the fact that he was born during Mardi Gras. He was only partly joking, since the instability of his character was real enough, oscillating as it did between elation and depression. He was short and stout, with dark, curly hair and the beard of a Taliban; his girlfriend called him “my dear sweet chubby one.” It was he, this Ludwig, who was the luckless hero of time’s directionality.

Sadi Carnot thought that heat was a substance, a fluid. He was wrong. Heat is the microscopic agitation of